

SIT281 Matrix Algebra

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score
Question 1	10 / 10
Question 2	3 / 6
Question 3	14 / 14
Question 4	14 / 14
Total	41 / 44 (93%)

Performance Summary

Exam Name:	SIT281 Matrix Algebra
Session ID:	0796322395
Exam Start:	Sat Jul 25 2026 10:34:16
Exam Stop:	Sat Jul 25 2026 11:40:54
Time Spent:	1:06:37

Question 1

Calculate the following (use the controls at the top to set the size of the matrix)

Rows: Columns:

$$3 \begin{pmatrix} -4 & 3 \\ -2 & 4 \end{pmatrix} - 5 \begin{pmatrix} 5 & -3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} -37 & 24 \\ -16 & -8 \end{pmatrix}$$

Expected answer: $\begin{pmatrix} -37 & 24 \\ -16 & -8 \end{pmatrix}$

Rows: Columns:

$$3 \begin{pmatrix} -4 & 3 \\ -2 & 4 \end{pmatrix} - 5 \begin{pmatrix} 5 & -3 \\ 2 & 4 \end{pmatrix} \pmod{3} = \begin{pmatrix} 2 & 0 \\ 2 & 1 \end{pmatrix}$$

Expected answer: $\begin{pmatrix} 2 & 0 \\ 2 & 1 \end{pmatrix}$

C

Your answer is correct. You were awarded 5 marks.

modC

Your answer is correct. You were awarded 5 marks.

You scored 10 marks for this part.

Score: 10/10

Question 2

Let $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 0 \\ 1 & 2 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 1 & 0 & 2 \end{pmatrix}$.

Compute $AB =$

Rows:	4	Columns:	3
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 $\begin{pmatrix} 4 & 3 & 8 \\ 3 & 3 & 6 \\ 1 & 1 & 2 \\ 3 & 3 & 6 \end{pmatrix}$ ✓ Expected answer: $\begin{pmatrix} 4 & 3 & 8 \\ 3 & 3 & 6 \\ 1 & 1 & 2 \\ 3 & 3 & 6 \end{pmatrix}$

Compute $AB \pmod{3} =$

Rows:	4	Columns:	3
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 $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ ✗ Expected answer: $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 1 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$

ABYour answer is correct. You were awarded **3** marks.**AB mod p**

Your answer is incorrect.

You scored **3** marks for this part.**Score: 3/**

Advice

To compute the product, we work out "row i of first matrix times column j of second matrix" for each entry i, j of the product. So in genera

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} = \begin{pmatrix} a_{11} \cdot b_{11} + a_{12} \cdot b_{21} + a_{13} \cdot b_{31} & a_{11} \cdot b_{12} + a_{12} \cdot b_{22} + a_{13} \cdot b_{32} & a_{11} \cdot b_{13} + a_{12} \cdot b_{23} + a_{13} \cdot b_{33} \\ a_{21} \cdot b_{11} + a_{22} \cdot b_{21} + a_{23} \cdot b_{31} & a_{21} \cdot b_{12} + a_{22} \cdot b_{22} + a_{23} \cdot b_{32} & a_{21} \cdot b_{13} + a_{22} \cdot b_{23} + a_{23} \cdot b_{33} \\ a_{31} \cdot b_{11} + a_{32} \cdot b_{21} + a_{33} \cdot b_{31} & a_{31} \cdot b_{12} + a_{32} \cdot b_{22} + a_{33} \cdot b_{32} & a_{31} \cdot b_{13} + a_{32} \cdot b_{23} + a_{33} \cdot b_{33} \\ a_{41} \cdot b_{11} + a_{42} \cdot b_{21} + a_{43} \cdot b_{31} & a_{41} \cdot b_{12} + a_{42} \cdot b_{22} + a_{43} \cdot b_{32} & a_{41} \cdot b_{13} + a_{42} \cdot b_{23} + a_{43} \cdot b_{33} \end{pmatrix}$$

So we have:

$$AB = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 0 \\ 1 & 2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 1 & 0 & 2 \end{pmatrix} =$$

$$\begin{pmatrix} 2 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 & 2 \cdot 1 + 1 \cdot 1 + 1 \cdot 0 & 2 \cdot 2 + 1 \cdot 2 + 1 \cdot 2 \\ 1 \cdot 1 + 2 \cdot 1 + 0 \cdot 1 & 1 \cdot 1 + 2 \cdot 1 + 0 \cdot 0 & 1 \cdot 2 + 2 \cdot 2 + 0 \cdot 2 \\ 0 \cdot 1 + 1 \cdot 1 + 0 \cdot 1 & 0 \cdot 1 + 1 \cdot 1 + 0 \cdot 0 & 0 \cdot 2 + 1 \cdot 2 + 0 \cdot 2 \\ 1 \cdot 1 + 2 \cdot 1 + 0 \cdot 1 & 1 \cdot 1 + 2 \cdot 1 + 0 \cdot 0 & 1 \cdot 2 + 2 \cdot 2 + 0 \cdot 2 \end{pmatrix} = \begin{pmatrix} 4 & 3 & 8 \\ 3 & 3 & 6 \\ 1 & 1 & 2 \\ 3 & 3 & 6 \end{pmatrix}.$$

Question 3

Consider the matrix

$$A = \begin{pmatrix} 4 & -2 & 2 \\ 1 & -1 & -4 \\ -4 & -3 & 4 \end{pmatrix}$$

Evaluate the determinant by expanding along row 3.

$$\begin{aligned} \det(A) &= \boxed{-4} \checkmark \quad \text{Expected answer: } \underline{-4} \times \det \begin{pmatrix} \boxed{-2} & \boxed{2} \\ \boxed{-1} & \boxed{-4} \end{pmatrix} \checkmark \quad \text{Expected answer: } \begin{pmatrix} \underline{-2} & \underline{2} \\ \underline{-1} & \underline{-4} \end{pmatrix} + \\ &\boxed{3} \checkmark \quad \text{Expected answer: } \underline{3} \times \det \begin{pmatrix} \boxed{4} & \boxed{2} \\ \boxed{1} & \boxed{-4} \end{pmatrix} \checkmark \quad \text{Expected answer: } \begin{pmatrix} \underline{4} & \underline{2} \\ \underline{1} & \underline{-4} \end{pmatrix} + \\ &\boxed{4} \checkmark \quad \text{Expected answer: } \underline{4} \times \det \begin{pmatrix} \boxed{4} & \boxed{-2} \\ \boxed{1} & \boxed{-1} \end{pmatrix} \checkmark \quad \text{Expected answer: } \begin{pmatrix} \underline{4} & \underline{-2} \\ \underline{1} & \underline{-1} \end{pmatrix} \\ &= \boxed{-4} \checkmark \quad \text{Expected answer: } \underline{-4} \times \boxed{10} \checkmark \quad \text{Expected answer: } \underline{10} + \\ &\boxed{3} \checkmark \quad \text{Expected answer: } \underline{3} \times \boxed{-18} \checkmark \quad \text{Expected answer: } \underline{-18} + \boxed{4} \checkmark \quad \text{Expected answer: } \underline{4} > \\ &\boxed{-2} \checkmark \quad \text{Expected answer: } \underline{-2} \\ &= \boxed{-102} \checkmark \quad \text{Expected answer: } \underline{-102} \end{aligned}$$

Hence,

$$\det(A) \pmod{18} = \boxed{6} \checkmark \quad \text{Expected answer: } \underline{6}$$

Note: If the coefficient of any sub-matrix determinant is zero, you will still need to include the submatrix in the first line of working and in its determinant in the second line of working.

Gap 0

Your answer is correct. You were awarded **1** mark.

Gap 1

Your answer is correct. You were awarded **1** mark.

Gap 2Your answer is correct. You were awarded **1** mark.**Gap 3**Your answer is correct. You were awarded **1** mark.**Gap 4**Your answer is correct. You were awarded **1** mark.**Gap 5**Your answer is correct. You were awarded **1** mark.**Gap 6**Your answer is correct. You were awarded **1** mark.**Gap 7**Your answer is correct. You were awarded **1** mark.**Gap 8**Your answer is correct. You were awarded **1** mark.**Gap 9**Your answer is correct. You were awarded **1** mark.**Gap 10**Your answer is correct. You were awarded **1** mark.**Gap 11**Your answer is correct. You were awarded **1** mark.**Gap 12**Your answer is correct. You were awarded **1** mark.**dmodp**Your answer is correct. You were awarded **1** mark.You scored **14** marks for this part.

Score: 14/1

Question 4

For the following matrix,

$$A = \begin{pmatrix} 8 & 9 & 3 \\ 5 & 5 & -2 \\ -6 & 0 & -7 \end{pmatrix}$$

First, calculate the cofactors of A ,

$c_{11} =$	$c_{12} =$	$c_{13} =$
-35 ✓ Expected answer: <u>-35</u>	47 ✓ Expected answer: <u>47</u>	30 ✓ Expected answer: <u>30</u>
$c_{21} =$	$c_{22} =$	$c_{23} =$
63 ✓ Expected answer: <u>63</u>	-38 ✓ Expected answer: <u>-38</u>	-54 ✓ Expected answer: <u>-54</u>
$c_{31} =$	$c_{32} =$	$c_{33} =$
-33 ✓ Expected answer: <u>-33</u>	31 ✓ Expected answer: <u>31</u>	-5 ✓ Expected answer: <u>-5</u>

Hence give the adjugate of A ,

Rows: Columns:

$\text{adj } A = \begin{pmatrix} \begin{matrix} \text{[-35]} & \text{[63]} & \text{[-33]} \\ \text{[47]} & \text{[-38]} & \text{[31]} \\ \text{[30]} & \text{[-54]} & \text{[-5]} \end{matrix} \end{pmatrix}$ ✓

Expected answer: $\begin{pmatrix} \begin{matrix} \text{[-35]} & \text{[63]} & \text{[-33]} \\ \text{[47]} & \text{[-38]} & \text{[31]} \\ \text{[30]} & \text{[-54]} & \text{[-5]} \end{matrix} \end{pmatrix}$

We have: $\det A =$ ✓ Expected answer: 233, and therefore

Rows: Columns:

$A^{-1} = \begin{pmatrix} \begin{matrix} \text{[-0.15]} & \text{[0.27]} & \text{[-0.14]} \\ \text{[0.20]} & \text{[-0.16]} & \text{[0.13]} \\ \text{[0.13]} & \text{[-0.23]} & \text{[-0.02]} \end{matrix} \end{pmatrix}$ ✓

Expected answer: $\begin{pmatrix} \begin{matrix} \text{[-0.15]} & \text{[0.27]} & \text{[-0.14]} \\ \text{[0.20]} & \text{[-0.16]} & \text{[0.13]} \\ \text{[0.13]} & \text{[-0.23]} & \text{[-0.02]} \end{matrix} \end{pmatrix}$ (enter your solution with two decimal values precision).

Finally, $(\det A)^{-1} \pmod{2} =$ ✓ Expected answer: 1, and so, in mod 2 arithmetic, the inverse of A is:

Rows: Columns:

$A^{-1} \pmod{2} = \begin{pmatrix} \begin{matrix} \text{[1]} & \text{[1]} & \text{[1]} \\ \text{[1]} & \text{[0]} & \text{[1]} \\ \text{[0]} & \text{[0]} & \text{[1]} \end{matrix} \end{pmatrix}$ ✓

Expected answer: $\begin{pmatrix} \begin{matrix} \text{[1]} & \text{[1]} & \text{[1]} \\ \text{[1]} & \text{[0]} & \text{[1]} \\ \text{[0]} & \text{[0]} & \text{[1]} \end{matrix} \end{pmatrix}$

Gap 0

Your answer is correct. You were awarded **1** mark.

Gap 1

Your answer is correct. You were awarded **1** mark.

Gap 2

Your answer is correct. You were awarded **1** mark.

Gap 3

Your answer is correct. You were awarded **1** mark.

Gap 4

Your answer is correct. You were awarded **1** mark.

Gap 5

Your answer is correct. You were awarded **1** mark.

Gap 6

Your answer is correct. You were awarded **1** mark.

Gap 7

Your answer is correct. You were awarded **1** mark.

Gap 8

Your answer is correct. You were awarded **1** mark.

Gap 9

Your answer is correct. You were awarded **1** mark.

deta

Your answer is correct. You were awarded **1** mark.

Ainv

Your answer is correct. You were awarded **1** mark.

Ainvmodp

Your answer is correct. You were awarded **1** mark.

invdeta

Your answer is correct. You were awarded **1** mark.

You scored **14** marks for this part.

Score: 14/1

Advice

Recall that to calculate the cofactors, c_{ij} we find the determinant of the matrix obtained by removing the i -th row and j -th column, and then into account the plus or minus. So here we have

$$c_{11} = \det \begin{pmatrix} 5 & -2 \\ 0 & -7 \end{pmatrix} = -35, \quad c_{12} = -\det \begin{pmatrix} 5 & -2 \\ -6 & -7 \end{pmatrix} = 47, \quad c_{13} = \det \begin{pmatrix} 5 & 5 \\ -6 & 0 \end{pmatrix} = 30,$$

$$c_{21} = -\det \begin{pmatrix} 9 & 3 \\ 0 & -7 \end{pmatrix} = 63, \quad c_{22} = \det \begin{pmatrix} 8 & 3 \\ -6 & -7 \end{pmatrix} = -38, \quad c_{23} = -\det \begin{pmatrix} 8 & 9 \\ -6 & 0 \end{pmatrix} = -54,$$

$$c_{31} = \det \begin{pmatrix} 9 & 3 \\ 5 & -2 \end{pmatrix} = -33, \quad c_{32} = -\det \begin{pmatrix} 8 & 3 \\ 5 & -2 \end{pmatrix} = 31, \quad c_{33} = \det \begin{pmatrix} 8 & 9 \\ 5 & 5 \end{pmatrix} = -5,$$

This hence gives the cofactor matrix,

$$\begin{pmatrix} -35 & 47 & 30 \\ 63 & -38 & -54 \\ -33 & 31 & -5 \end{pmatrix}$$

Then we take the transpose to find the adjugate,

$$\text{adj}A = \begin{pmatrix} -35 & 63 & -33 \\ 47 & -38 & 31 \\ 30 & -54 & -5 \end{pmatrix}$$

We should be able to verify that multiplying the original matrix and the adjugate produces a diagonal matrix with entries equal to the determinant of A . In this case

$$\begin{pmatrix} 8 & 9 & 3 \\ 5 & 5 & -2 \\ -6 & 0 & -7 \end{pmatrix} \begin{pmatrix} -35 & 63 & -33 \\ 47 & -38 & 31 \\ 30 & -54 & -5 \end{pmatrix} = \begin{pmatrix} 233 & 0 & 0 \\ 0 & 233 & 0 \\ 0 & 0 & 233 \end{pmatrix}$$

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(http://www.newcastle.ac.uk\)](http://www.newcastle.ac.uk).

Self Assessment Quiz

Question ①

$$\begin{aligned} \textcircled{1} \quad 3 \begin{bmatrix} -4 & 3 \\ -2 & 4 \end{bmatrix} - 5 \begin{bmatrix} 5 & -3 \\ 2 & 4 \end{bmatrix} &= \begin{bmatrix} -12 & 9 \\ -6 & 12 \end{bmatrix} - \begin{bmatrix} 25 & -15 \\ 10 & 20 \end{bmatrix} \\ &= \begin{bmatrix} -12-25 & 9+15 \\ -6-10 & 12+20 \end{bmatrix} \\ &= \begin{bmatrix} -37 & 24 \\ -16 & -8 \end{bmatrix} \end{aligned}$$

$$\begin{bmatrix} -37 & 24 \\ -16 & -8 \end{bmatrix} \pmod{3} = \begin{bmatrix} 2 & 0 \\ 2 & 1 \end{bmatrix}$$

Question 2

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 1 & 0 & 2 \end{bmatrix}$$

$$2 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 = 4$$

$$1 \cdot 1 + 2 \cdot 1 + 1 \cdot 0 = 3$$

$$2 \cdot 1 + 1 \cdot 1 + 0 = 3$$

$$1 \cdot 1 + 2 \cdot 1 + 0 \cdot 0 = 3$$

$$2 \cdot 2 + 1 \cdot 2 + 1 \cdot 2 = 8$$

$$1 \cdot 2 + 2 \cdot 2 + 0 = 6$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 0 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 4 & 3 & 8 \\ 3 & 3 & 6 \\ 1 & 1 & 2 \\ 3 & 3 & 6 \end{bmatrix}$$

$$0 + 1 \cdot 1 + 0 = 1$$

$$1 \cdot 2 + 2 \cdot 2 + 0 = 6$$

$$0 + 1 \cdot 1 + 0 = 1$$

$$1 \cdot 1 + 2 \cdot 1 + 0 = 3$$

$$0 + 1 \cdot 2 + 0 = 2$$

$$1 \cdot 1 + 2 \cdot 1 + 0 = 3$$

$$AB = \begin{bmatrix} 4 & 3 & 8 \\ 3 & 3 & 6 \\ 1 & 1 & 2 \\ 3 & 3 & 6 \end{bmatrix}$$

$$AB \pmod{3} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 1 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

!

I've accidentally written this as 1 in NUMBAS.

Question 3

$$\textcircled{3} \quad A = \begin{bmatrix} 4 & -2 & 2 \\ 1 & -1 & -4 \\ -4 & -3 & 4 \end{bmatrix}$$

$$\begin{aligned} \det(A) &= -4 \times C_{31} + -3 \times C_{32} + 4 \times C_{33} \\ &= -4 \times 1 \times \det \begin{pmatrix} -2 & 2 \\ -1 & -4 \end{pmatrix} + -3 \times -1 \times \det \begin{pmatrix} 4 & 2 \\ 1 & -4 \end{pmatrix} + 4 \times 1 \times \det \begin{pmatrix} 4 & -2 \\ 1 & -1 \end{pmatrix} \\ &= -4(8+2) + 3(-16-2) + 4(-4+2) \\ &= -4 \times 10 + 3 \times -18 + 4 \times -2 \\ &= -102 \end{aligned}$$

$$-102 \equiv 6 \pmod{18}$$

$$18 \times 6 = 102 \Rightarrow 6$$

Question ④

$$A = \begin{pmatrix} 8 & 9 & 3 \\ 5 & 5 & -2 \\ -6 & 0 & -7 \end{pmatrix}$$

$$C_{11} = (-1)^{1+1} \times \det \begin{pmatrix} 5 & -2 \\ 0 & -7 \end{pmatrix} = 1 \times (-35 + 0) = -35$$

$$C_{12} = (-1)^{1+2} \times \det \begin{pmatrix} 5 & -2 \\ -6 & -7 \end{pmatrix} = -1 \times (5 \times -7 - 6 \times 2) = 47$$

$$C_{13} = (-1)^{1+3} \times \det \begin{pmatrix} 5 & 5 \\ -6 & 0 \end{pmatrix} = 1 \times (0 + 5 \times 6) = 30$$

$$C_{21} = (-1)^{2+1} \times \det \begin{pmatrix} 9 & 3 \\ 0 & -7 \end{pmatrix} = -1 \times (9 \times -7 - 0) = 63$$

$$C_{22} = (-1)^{2+2} \times \det \begin{pmatrix} 8 & 3 \\ -6 & -7 \end{pmatrix} = 1 \times (8 \times -7 + 6 \times 3) = -38$$

$$C_{23} = (-1)^{2+3} \times \det \begin{pmatrix} 8 & 9 \\ -6 & 0 \end{pmatrix} = -1 \times (8 \times 0 + 9 \times 6) = -54$$

$$C_{31} = (-1)^{3+1} \times \det \begin{pmatrix} 9 & 3 \\ 5 & -2 \end{pmatrix} = 1 \times (9 \times -2 - 15) = -33$$

$$C_{32} = (-1)^{3+2} \times \det \begin{pmatrix} 8 & 3 \\ 5 & -2 \end{pmatrix} = -1 \times (-16 - 15) = 31$$

$$C_{33} = (-1)^{3+3} \times \det \begin{pmatrix} 8 & 9 \\ 5 & 5 \end{pmatrix} = +1 \times (8 \times 5 - 9 \times 5) = -5$$

$$\therefore C(A) = \begin{bmatrix} -35 & 47 & 30 \\ 63 & -38 & -54 \\ -33 & 31 & -5 \end{bmatrix}$$

$$\det(A) = 8 \times C_{11} + 9 \times C_{12} + 3 \times C_{13}$$

$$= 8 \times -35 + 9 \times 47 + 3 \times 30$$

$$= 233$$

$$\text{adj } A = \begin{bmatrix} -35 & 63 & -33 \\ 47 & -38 & 31 \\ 30 & -54 & -5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{233} \text{adj } A$$

$$A^{-1} = \frac{1}{233} \begin{bmatrix} -35 & 63 & -33 \\ 47 & -38 & 31 \\ 30 & -54 & -5 \end{bmatrix} = \begin{bmatrix} -35/233 & 63/233 & -33/233 \\ 47/233 & -38/233 & 31/233 \\ 30/233 & -54/233 & -5/233 \end{bmatrix} = \begin{bmatrix} -0.15 & 0.27 & -0.14 \\ 0.2 & -0.16 & 0.13 \\ 0.13 & -0.23 & -0.23 \end{bmatrix}$$

$$233 \equiv 1 \pmod{2} \Rightarrow (\det A)^{-1} \pmod{2} = 1$$

$$A^{-1} = \frac{1}{233} \begin{bmatrix} -35 & 63 & -33 \\ 47 & -38 & 31 \\ 30 & -54 & -5 \end{bmatrix} \equiv 1 \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \pmod{2}$$

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